

Здание 17 - по ядру Linux

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1) Настраиваем grub

```
GNU nano 6.2 /etc/default/grub
# If you change this file, run 'update-grub' afterwards to update
# /boot/grub/grub.cfg.
# For full documentation of the options in this file, see:
#   info -f grub -n 'Simple configuration'

GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=menu
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
GRUB_CMDLINE_LINUX=""

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xefefefef"

# Uncomment to disable graphical terminal (grub-pc only)
#GRUB_TERMINAL=console

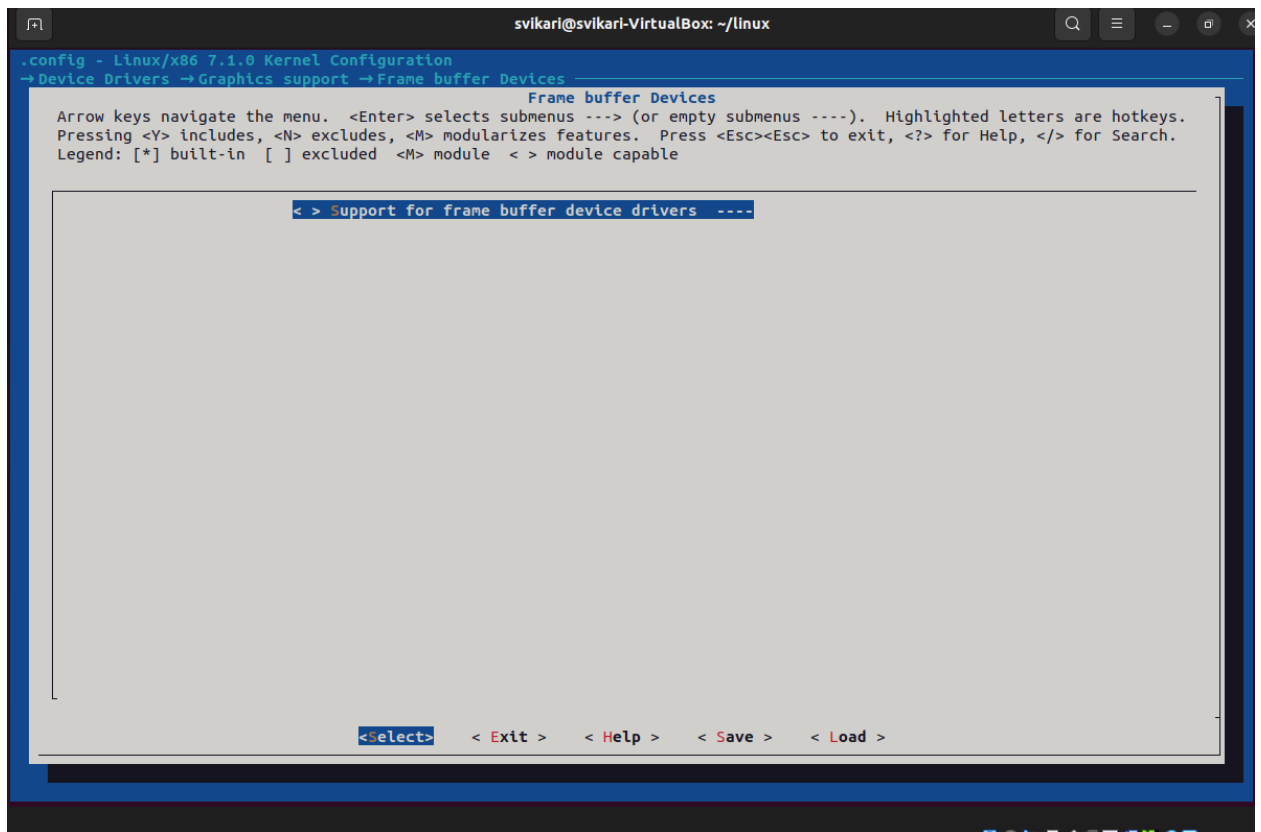
[ Wrote 33 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

2) Клонировем linux, после чего собираем с помощью make defconfig

```
svikari@svikari-VirtualBox:~$ git clone https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git
Cloning into 'linux'...
remote: Enumerating objects: 11623749, done.
remote: Counting objects: 100% (3168/3168), done.
remote: Compressing objects: 100% (1351/1351), done.
remote: Total 11623749 (delta 2292), reused 2198 (delta 1799), pack-reused 11620581 (from 1)
Receiving objects: 100% (11623749/11623749), 3.19 GiB | 10.12 MiB/s, done.
Resolving deltas: 100% (9565195/9565195), done.
Updating files: 100% (94326/94326), done.

svikari@svikari-VirtualBox:~/linux$ make defconfig
LEX      scripts/kconfig/lexer.lex.c
YACC     scripts/kconfig/parser.tab.[ch]
HOSTCC   scripts/kconfig/lexer.lex.o
HOSTCC   scripts/kconfig/menu.o
HOSTCC   scripts/kconfig/parser.tab.o
HOSTCC   scripts/kconfig/preprocess.o
HOSTCC   scripts/kconfig/symbol.o
HOSTCC   scripts/kconfig/util.o
HOSTLD   scripts/kconfig/conf
*** Default configuration is based on 'x86_64_defconfig'
#
# configuration written to .config
#
svikari@svikari-VirtualBox:~/linux$
```

3) С помощью menuconfig смотрим открывшееся меню



- 4) Выполняем `make bzImage`, `make modules`, `sudo make modules_install`, `sudo make install` для сборки ядра, после чего перезапускаем систему и смотрим версию ядра.

```

CC      fs/nfs/callback.o
CC      fs/nfs/callback_xdr.o
CC      fs/nfs/callback_proc.o
CC      fs/nfs/nfs4namespace.o
Rhythmbox fs/nfs/nfs4getroot.o
CC      fs/nfs/nfs4client.o
CC      fs/nfs/nfs4session.o
CC      fs/nfs/dns_resolve.o
CC      fs/nfs/nfs4trace.o
CC      fs/nfs/pnfs.o
CC      fs/nfs/pnfs_dev.o
CC      fs/nfs/pnfs_nfs.o
CC      fs/nfs/nfs4sysctl.o
CC      fs/nfs/nfs40client.o
CC      fs/nfs/nfs40proc.o
AR      fs/nfs/built-in.a
CC      fs/exportfs/expfs.o
AR      fs/exportfs/built-in.a
CC      fs/lockd/clntlock.o
CC      fs/lockd/clntproc.o
CC      fs/lockd/clntxdr.o
CC      fs/lockd/host.o
CC      fs/lockd/svc.o
CC      fs/lockd/svclock.o
CC      fs/lockd/svcshare.o
CC      fs/lockd/svcproc.o
CC      fs/lockd/svcsubs.o
CC      fs/lockd/mon.o
CC      fs/lockd/trace.o
CC      fs/lockd/xdr.o
CC      fs/lockd/netlink.o
CC      fs/lockd/clnt4xdr.o
CC      fs/lockd/svc4proc.o

```

```

CC      arch/x86/boot/video-vga.o
CC      arch/x86/boot/video-vesa.o
CC      arch/x86/boot/video-bios.o
LD      arch/x86/boot/setup.elf
OBJCOPY arch/x86/boot/setup.bin
OBJCOPY arch/x86/boot/vmlinux.bin
BUILD   arch/x86/boot/bzImage
Kernel: arch/x86/boot/bzImage is ready (#1)
svikari@svikari-VirtualBox:~/linux$
svikari@svikari-VirtualBox:~/linux$ make modules
DESCEND objtool
CC [M]  fs/efivarfs/inode.o
CC [M]  fs/efivarfs/file.o
CC [M]  fs/efivarfs/super.o
CC [M]  fs/efivarfs/vars.o

```

```
svikari@svikari-VirtualBox:~/linux$ sudo make modules_install
[sudo] password for svikari:
SYMLINK /lib/modules/7.1.0-07602-g66affa37cfac/build
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/modules.order
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/modules.builtin
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/modules.builtin.modinfo
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/fs/efivarfs/efivarfs.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/drivers/thermal/intel/x86_pkg_temp_thermal.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/netfilter/nf_log_syslog.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/netfilter/xt_mark.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/netfilter/xt_LOG.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/netfilter/xt_MASQUERADE.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/netfilter/xt_addrtype.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/ipv4/netfilter/nf_reject_ipv4.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/ipv4/ip_tunnel.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/ipv4/tunnel4.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/ipv6/netfilter/nf_reject_ipv6.ko
INSTALL /lib/modules/7.1.0-07602-g66affa37cfac/kernel/net/ipv6/sit.ko
DEPMOD /lib/modules/7.1.0-07602-g66affa37cfac
```

```
svikari@svikari-VirtualBox:~$ uname -r
7.1.0-07602-g66affa37cfac
svikari@svikari-VirtualBox:~$
```